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# Complex integration

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(1)

Recall that an analytic function has derivatives of all orders. This is due to Cauchy's integral formula.

Applications of complex integration : problems in physics  
special functions like gamma fn & error function.

In real analysis / calculus, we study

indefinite integrals  $\int_{x_0}^x f(t) dt$

Integrating over  $\begin{array}{c} + \text{---} + \\ x_0 \quad x \end{array}$   
or

definite integrals  $\int_a^b f(t) dt$

$\begin{array}{c} + \text{---} + \\ a \quad b \end{array}$

Here, in complex analysis, there is not one path between  $x_0$  &  $x$ . We study complex integrals along a given contour (between  $a$  &  $b$  or  $x_0$  &  $x$ ) & these are called contour integrals or line integrals.

## Review of definitions

1) Arc/path/curve : A set of points  $z = x + iy$  in the complex plane is said to be a curve if  $x = x(t)$  &  $y = y(t)$  are functions of  $t$  where  $a \leq t \leq b$  &  $x(t)$  &  $y(t)$  are continuous wrt  $t$ .

If we call this curve as  $C$ , then  $z(t) = x(t) + iy(t)$  is called a parametric representation of  $C$ .

2) Eg 1) The unit circle Simple closed curve : If a curve is such that only the initial & final values of  $c(t)$  are the same, then such curves are called simple closed curves.

Ex 1) The unit circle  $z = e^{i\theta}$ ,  $0 \leq \theta \leq 2\pi$  about the origin  
is a simple closed curve with ~~the~~ a parametrisation

$$z = e^{i\theta}, \quad 0 \leq \theta \leq 2\pi$$

It also has another parametrisation

3) A curve ~~C~~ or  $z(t)$ ,  $0 \leq t \leq 2\pi$

3) A curve  $C$  or  $z(t)$ ,  $a \leq t \leq b$  is said to be a smooth curve if  $(C \neq \text{smooth})$

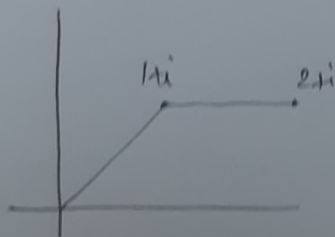
a)  $\frac{dz}{dt}$  exists & is continuous on  $[a, b]$ , &

b)  $\frac{dz}{dt} \neq 0$  throughout  $(a, b)$

4) A contour or piecewise smooth curve is a curve consisting of a finite number of smooth curves joined end to end.

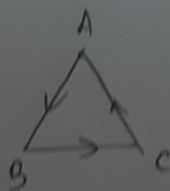
Ex. The polygonal line

$$z = \begin{cases} x + iz & 0 \leq x \leq 1 \\ x + i & 1 \leq x \leq 2 \end{cases}$$

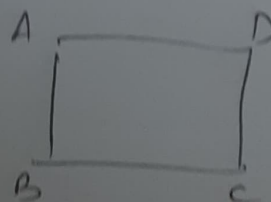


$\therefore$  if  $z(t)$ ,  $a \leq t \leq b$  represents a contour, then  $z(t)$  is continuous on  $[a, b]$  &  $z'(t)$  is piecewise continuous.

Simple closed curves



or

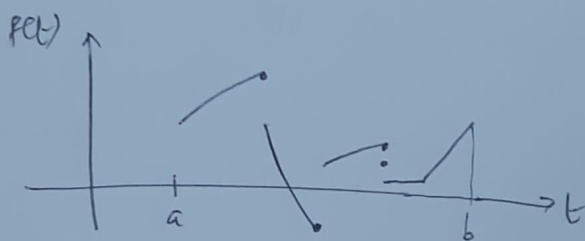


## Piecewise cts on $[a, b]$

$f$  is said to be piecewise cts on  $[a, b]$  if it is continuous everywhere on  $[a, b]$  except possibly for a finitely many points where although discontinuous, it has one sided limits.

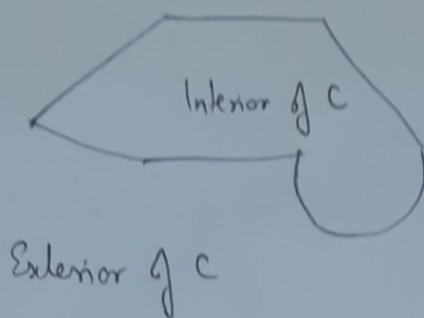
## Piecewise smooth on $[a, b]$

$f$  is said to be piecewise smooth on  $[a, b]$  if it is  $C^1$  smooth everywhere on  $[a, b]$  except possibly for finitely many pts where ~~although~~ derivatives exist, they may not be continuous.



5) Interior & exterior of a simple closed contour C

(3)



Interior of C is bounded & (Jordan's curve theorem)  
exterior of C is unbounded

Points on C are the boundaries of both these domains.

Contour Integral

Given a contour C parametrised by  $z: [a, b] \rightarrow \mathbb{C}$  &  $f$  a continuous function on C, we define the contour integral / line integral / integral of  $f$  along C as

$$\int_C f(z) dz := \int_a^b f(z(t)) \frac{dz(t)}{dt} dt$$

Note that a contour may have more than one parametrisation, however, integral value is independent of the parametrisation.

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Properties of Contour Integrals

1)  $\int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz$

(Integration is linear)

2) (Sense reversal)  $\int_{z_0}^z f(z) dz = - \int_z^{z_0} f(z) dz$

### 3) Partitioning of path

(4)

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$



A domain  $D$  is called simply connected if every simple closed curve within it encloses only points of  $D$ .

Eg Open circular disc

$$B(a, r)$$

Non eg.

Open Annulus

$$r_1 < |z-a| < r_2$$

### Evaluating Contour Integrals

There are two methods of evaluating contour integrals.

Method 1) Using "antiderivative." (will be explained later)

$$\text{Eg 5)} \int_0^{1+i} z^2 dz = \left. \frac{z^3}{3} \right|_0^{1+i} = \frac{(1+i)^3}{3}$$

(Here, we used the fact that  $z^2$  has an antiderivative  $\frac{z^3}{3}$ .)

The following theorem guarantees the existence of "antiderivatives" in  $SCD_s$ .

Theorem (Kreyszig Pg 647 Section 14.1)

Let  $f(z)$  be analytic in a simply connected domain  $D$ . Then,  $\exists$  an

a) analytical function  $F(z)$  in  $D$  such that

$$F'(z) = f(z) \text{ on } D$$

(This will be called the indefinite integral of  $f$  in  $D$ ) & (5)  
 Let  $z_0$  &  $z_1$  be 2 pts on  $D$  & let  $C$  be any path with IP  $z_0$  & EP  $z_1$ .  
 b) ~~for any path  $C$  in  $D$ , joining two points  $z_0$  &  $z_1$  in  $D$ , we have~~  
~~(oriented from  $z_0$  to  $z_1$ )~~  

$$\int_C f(z) dz = F(z_1) - F(z_0) \quad (\text{inside } D)$$

Remark :- Since integral is indep't of the path chosen from  $z_0$  to  $z_1$ , we can as well write it uniformly as  $\int_{z_0}^{z_1} f(z) dz$ .

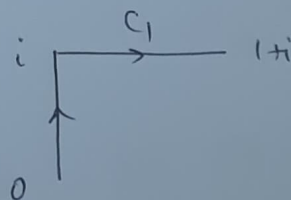
Eg 1)  $\int_{-Ti}^{Ti} \cos z \, dz = 2 \sin(Ti)$

2)  $\int_{8+Ti}^{8-3Ti} e^{z/2} \, dz = 0$

Method 2: Using parametrisation

Find the contour integral a)  $\int_{C_1} f(z) dz$  where (Eg 2: Churchi 1)

$f(z) = y - x - i3x^2$



b)  $\int_{C_2} f(z) dz$





a)

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Note that if we do integration on simple closed curves, there is an ambiguity in the ~~de~~ orientation. In such cases, we often assume (if not explicitly given otherwise) that orientation is in the direction along which parameter  $t$  increases.

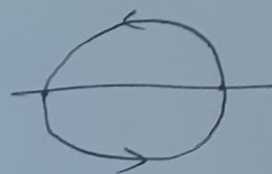
for example if

2)  $\oint_C \frac{dz}{z}$  where  $C$  is the unit circle oriented in the counter clockwise direction parametrised as  $z(t) = \cos t + i \sin t$   $-\pi \leq t \leq \pi$  (7)

If nothing is specified, it is assumed that it is counter clockwise as  $t$  as  $t$  increases,  $z(t)$  traverses Ccw.

2) b)  $\oint_C \frac{dz}{z}$  . Let  $z = e^{it}$ ,  $-\pi \leq t \leq \pi$

$$\int_{-\pi}^{\pi} \frac{1}{e^{it}} i e^{it} dt = 2\pi i$$



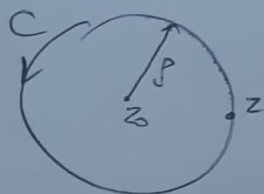
a)  $\oint_C z^2 dz = \int_{-\pi}^{\pi} e^{2it} i e^{it} dt = i \frac{e^{3it}}{3i} \Big|_{-\pi}^{\pi} = \frac{1}{3} \times [e^{3\pi i} - e^{-3\pi i}]$   
 $= \frac{1}{3} [e^{\pi i} - e^{\pi i}] = 0.$

Conclusion:  $\frac{1}{z}$  is not analytic on any SCD which contains  $C$ .

c)  $\int_{-\pi}^{\pi} e^{mx} dx$

d)  $\oint_C (z-z_0)^m dz$  where  $m \in \mathbb{Z}$  &  $z_0$  is a constant.

Integrate counter clockwise around the circle  $C$  of radius  $\rho$  with center at  $z_0$ .



$$z(t) - z_0 = \rho e^{it} \quad -\pi \leq t \leq \pi \text{ or } 0 \leq t \leq 2\pi$$

$C : z(t) = z_0 + \rho e^{it}$

$$\frac{dz}{dt} = \rho i e^{it}$$

$$\int_{-\pi}^{\pi} (\rho e^{it})^m i \rho e^{it} dt = i \rho^{m+1} \int_{-\pi}^{\pi} e^{i(m+1)t} dt$$



$$\int_{-\pi}^{\pi} e^{mxi} dx = \frac{e^{mxi}}{mi} \Big|_{-\pi}^{\pi} \quad \text{if } m \neq 0$$

$$= 2\pi \quad \text{if } m = 0$$

$$\& e^{m\pi i} = \cancel{e^{m\pi i}} = (e^{\pi i})^m = (-1)^m$$

$$\& e^{-m\pi i} = (e^{\pi i})^{-m} = (-1)^{-m}$$

both are same. Hence,

$$\int_{-\pi}^{\pi} e^{mxi} dx = \begin{cases} 0 & \text{if } m \neq 0 \\ 2\pi & \text{if } m = 0. \end{cases}$$

Exercise: If  $m, n \in \mathbb{Z}$ ,

$$\int_0^{2\pi} e^{im\theta} e^{-in\theta} d\theta = \begin{cases} 0 & \text{when } m \neq n \\ 2\pi & \text{when } m = n \end{cases}$$

Back to part (c) :-

$$= \begin{cases} 0 & \text{if } m \neq -1 \\ 2\pi i & \text{if } m = -1 \end{cases} = \begin{cases} 0 & \text{if } m \neq -1 \\ 2\pi i & \text{if } m = -1. \end{cases}$$

~~Suggests~~ <sup>could be</sup>  ~~$f(z)$  is analytic on some SCD containing~~

Since  $f(z) = (z-z_0)^m$  is analytic on any SCD containing  $C$ , this was expected, as integral depends on end pts only.

## ML inequality

(9)

Let  $C$  denote a ~~contour~~ smooth  $C^1$  smooth curve in  $\mathbb{C}$ .

say  $C: z(t)$ ,  $a \leq t \leq b$

ie,  $\frac{dz}{dt}(t)$  is ~~cts~~ <sup>non zero</sup> on  $[a, b]$  & ~~differentiable~~ on  $(a, b)$ .

Recall <sup>(arclength)</sup> length of a smooth curve was defined as

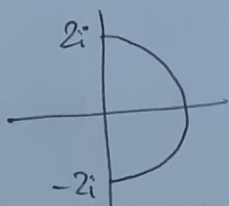
$$\text{len}(C) := \int_a^b \left| \frac{dz}{dt}(t) \right| dt$$

Now if  $C$  is a piecewise smooth curve or contour, then

~~say~~  $C$  can be written as union of  $n$  many smooth curves say  $C = \bigcup_{i=1}^n C_i$ , then

$$\text{len}(C) = \sum_{i=1}^n \text{len}(C_i)$$

Eg. Perimeter of a circle half circle (of radius 2)



$$z(t) = 2e^{it} \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

$$\text{len}(C) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |2ie^{it}| dt = 2\pi$$

## Bounds for contour integral

Many a times, there will be need for estimating the absolute value of contour integrals. For this, we have the following theorem:-

Theorem ~~called~~ (M-L inequality)

Let  $C$  denote a contour of length  $L$  & suppose that a function  $f(z)$  is piecewise continuous on  $C$ . If  $M$  is a non-negative constant such that  $|f(z)| \leq M$  for all points  $z$  on  $C$  at which  $f(z)$  is

defined, then

(10)

$$\left| \int_C f(z) dz \right| \leq ML.$$

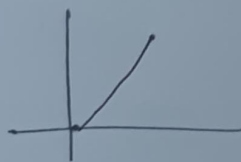
Pr 1) Find an upper bound for the absolute value of the integral  $\int_C z^2 dz$  where  $C$  is the straight line segment from 0 to  $1+i$ .

We had done this:  $\frac{(1+i)^3}{3}$

Find  $M$  s.t.  $|f(z)| \leq M$

$$|z^2| = |(x+iy)^2|$$

Parametrisation  $z(t) : t+it$   
 $0 \leq t \leq 1.$



[You may also write  $z(x) : x+ix$ ,  $0 \leq x \leq 1.$ ]

$$\therefore |(t+it)^2| = t^2 \times |(1+i)^2| = 2t^2 \leq 2$$

$M=2$  is an option (Any  $M \geq 2$  works)

So, now we find  $\text{len}(C)$  (obviously we know it is  $\sqrt{2}$ )

$$\text{len}(C) = \int_0^1 \left| \frac{dz(t)}{dt} \right| dt = \int_0^1 |1+i| dt = \sqrt{2}.$$

$$\therefore \left| \int_C f(z) dz \right| \leq 2\sqrt{2}.$$

Pr 2) Let  $C$  be the arc of the circle  $|z|=2$  from  $z=2$  to  $z=2i$ , that lies in the first quadrant. Show that

$$\left| \int_C \frac{z+4}{z^3-1} dz \right| \leq \frac{6\pi}{7}.$$

f satisfies the conditions of ML inequality theorem

What is the M

$$\text{st } |f(z)| \leq M \quad \forall z \in C.$$

$$|f(z)| = \left| \frac{z+4}{z^3-1} \right| \leq \frac{|z|+4}{|z^3-1|} = \frac{6}{|z^3-1|}$$

Lower bound for  $|z^3-1|$  is given by reverse triangle inequality

$$|a-b| \geq ||a| - |b||$$

Proof:- let  $a, b \in \mathbb{C}$ .

$$|a| = |a+b-b| \leq |a+b| + |-b| =$$

$$\Rightarrow |a+b| \geq |a| - |b|$$

$$\text{ii) } |b| = |b+a-a| \leq |a+b| + |a|$$

$$\Rightarrow |a+b| \geq |b| - |a|$$

Put together

$$|a+b| \geq ||a| - |b||$$

$$\text{Hence, } |z^3-1| \geq (|z|^3-1) = 8-1 = 7$$

$$\therefore \frac{1}{|z^3-1|} \leq \frac{1}{7}$$

$$\Rightarrow |f(z)| \leq \frac{6}{7}$$

$$\therefore \left( \int_C |f(z)| dz \right) \leq \frac{6}{7} \times L \quad \text{where}$$

$$L \text{ is } \frac{1}{4} (\text{perimeter of circle of radius 2}) = \frac{2\pi \times 2}{4} = \pi$$

$$\therefore \text{Ans : } \underline{\underline{\frac{6\pi}{7}}}$$

$$z(t) = 2e^{it}, \quad 0 \leq t \leq \frac{\pi}{2}$$

$$z'(t) = 2ie^{it}$$

$$\text{len}(C) = \int_0^{\frac{\pi}{2}} 2 \, dt = \pi$$



# (Kreyszig Pg 652) Cauchy's Integral Theorem

(1)

If  $f(z)$  is analytic in a simply connected domain  $D$ , then for every simple closed contour  $C$ , then

$$\oint_C f(z) dz = 0$$

Eg.  $\oint_C e^z dz = 0$  ;  $\oint_C \cos z dz = 0$  ;  $\oint_C z^n dz = 0$  ( $n=0,1,\dots$ )

For any ~~closed~~ contour  $C$  in  $D$

Pr. 1) Let  $C$  be the unit circle. Find a)  $\oint_C \sec z dz$

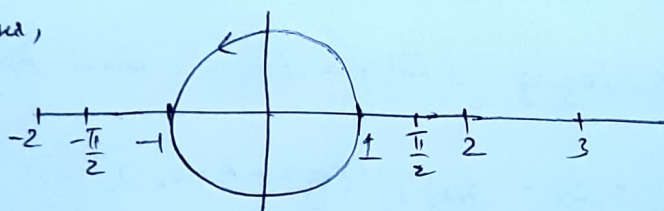
$\sec z = \frac{1}{\cos z}$  is not analytic when  $\cos z = 0$

ie, when  $z = \frac{\pi}{2} + n\pi$ ,  $n \in \mathbb{Z}$

ie,  $\left\{ -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \dots \right\}$

Skip this as trigonometric fns have not been introduced extensively

However,



all such points ~~are~~ can be brought outside the simply connected domain

$D = \left\{ z \in \mathbb{C} \mid |z| < 1 + \frac{\pi}{4} \right\}$  where ~~for~~  $\sec z$  is analytic

$$\Rightarrow \oint_C \sec z dz = 0$$

Do this

~~Ex. 1) a)~~ b)  $\oint_C \frac{1}{z^2+4} dz$

Pr. 2) c)  $\oint_C \bar{z} dz = \int_{-\pi}^{\pi} e^{-it} i e^{it} dt = i \times 2\pi$

d)  $\oint_C \frac{1}{z} dz = 2\pi i$

$$z(t) = e^{it}$$

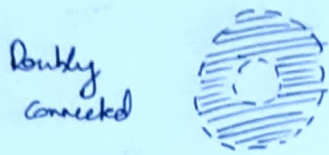
$$-\pi \leq t \leq \pi$$

$$\frac{dz}{dt} = i e^{it}$$

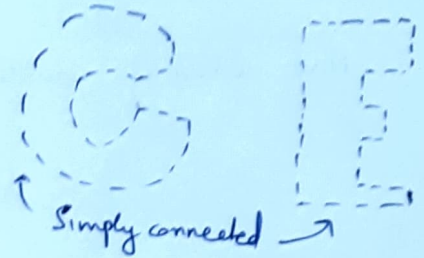
A domain that is not simply connected is called multiply connected.

Eg- Annulus : It is doubly connected ;  $1 < |z-a| < 2$

A disc without the circle, say,  $0 < |z| < 1$



Triply connected



Now let us discuss the Cauchy's Theorem

- 1) Analyticity
- 2) Simply connectedness of the domain
- 3) closed

Pr 2)  $\oint_C \frac{1}{z^2} dz$  with  $C = \text{Unit Circle}$

$\frac{1}{z^2}$  is not analytic on any RCD containing  $C$ . However,

$$\oint_C \frac{1}{z^2} dz = 0$$

$z(t) = e^{it}$  ;  $-\pi \leq t \leq \pi$        $\frac{dz}{dt} = ie^{it}$

$$\Rightarrow \oint_C \frac{1}{z^2} dz = \int_{-\pi}^{\pi} e^{-2it} ie^{it} dt = i \left. \frac{e^{-it}}{-i} \right|_{-\pi}^{\pi} = e^{\pi i} - e^{-\pi i} = -1 - (-1) = 0$$

To clarify,

$f$  analytic on  $D$   $\Rightarrow \oint_C f(z) dz = 0$   
 $\Delta C \subseteq D$

However if  $\exists$  a RCD  $D$  containing  $C$  such that

~~However~~  $\oint_C f(z) dz = 0$ ,  ~~$f \in \text{RCD}$~~   $f$  is analytic on  $D$  always (as in the above case)  
it doesn't imply that



(3)

Note that 1) Analyticity 2) Simply connectedness of the domain 3) "closed" contours are 3 key ingredients of the theorem. If we try to remove <sup>(1)</sup> them, integral may or may not be zero.

For eg. if we consider Pr 1 c), integral is not longer zero as  $\bar{z}$  is not analytic on ~~any~~ ~~set~~ ~~that~~ ~~contains~~  $C$ .

However, if we consider  $f(z) = \frac{1}{z^2}$ , then  $\oint \frac{1}{z^2} dz$  turns out to be zero even though  $\frac{1}{z^2}$  is not analytic on any set that contains  $C$ .

Calculation:  $\oint_C \frac{1}{z^2} dz$   $C: z(t) = -\pi \leq t \leq \pi, e^{it}$

$$= \int_{-\pi}^{\pi} e^{-2it} i e^{it} dt = i \left. \frac{e^{-it}}{-i} \right|_{-\pi}^{\pi} = e^{-it} \Big|_{\pi}^{-\pi} = e^{\pi i} - e^{-\pi i} = (-1) - (-1) = 0$$

### Simply connectedness (2) Consider Pr 1 d)

If we replace SCD by a domain that is doubly connected say

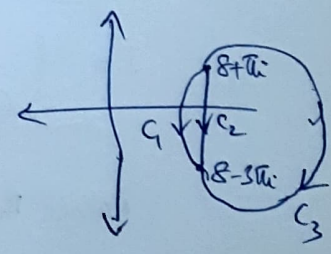
$A = \{ z \in \mathbb{C} \mid \frac{1}{2} < |z| < \frac{3}{2} \}$ , then  $f(z) = \frac{1}{z}$  is analytic on  $A$  but

$$\oint_C \frac{1}{z} dz = 2\pi i \text{ still holds.}$$

Simply connectedness is necessary (as to present a theorem where  $\oint f(z) dz = 0$ ).

### Closed (3) Let us consider any contour $C$ from

Recall  $\int_{8+\pi i}^{8-3\pi i} e^{z/2} dz = 2e^{z/2} \Big|_{8+\pi i}^{8-3\pi i} = 0$

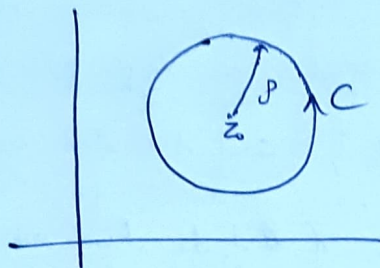


And so  $\int_{-\pi}^{\pi} e^{mit} dt = \begin{cases} \frac{e^{mit}}{mi} \Big|_{-\pi}^{\pi} = 0 & \text{if } m \neq 0 \\ 2\pi & \text{if } m = 0 \end{cases} = \begin{cases} \frac{e^{\pi im} - e^{-\pi im}}{mi} = (-1)^m - (-1)^m = 0 & \text{if } m \neq 0 \\ 2\pi & \text{if } m = 0 \end{cases}$

where  $m \in \mathbb{Z}$

Pr 2) Let  $f(z) = (z-z_0)^m$  where  $m \in \mathbb{Z}$  &  $z_0$  is a constant.

Integrate ccw around the circle  $C$  of radius  $\rho$  with center at  $z_0$ .



By CIT,  $f(z)$  is analytic on any SED that contains  $C$  as long as  $m \geq 0$ . Hence, clearly,

$$\oint_C (z-z_0)^m dz = 0 \quad \forall m \geq 0.$$

Case :  $m = -1$

$$\oint_C \frac{1}{z-z_0} dz$$

$$\text{Let } z(t) = z_0 + \rho e^{it}, \quad -\pi \leq t \leq \pi$$

$$\frac{dz}{dt} = i \rho e^{it}$$

$$\int_{-\pi}^{\pi} \frac{1}{\rho e^{it}} \times i \rho e^{it} dt = 2\pi i$$

Case :  $m \leq -2$  : let  $n = |m|$

$$I = \oint_C \frac{1}{(z-z_0)^n} dz = \int_{-\pi}^{\pi} \frac{1}{(\rho e^{it})^n} i \rho e^{it} dt = i \int_{-\pi}^{\pi} (\rho e^{it})^{-n+1} dt$$

Recall that  $\int_{-\pi}^{\pi} e^{imt} dt = \begin{cases} 0 & \text{if } m \neq 0 \\ 1 & \text{if } m = 0 \end{cases}$

$$\text{Hence, } I = i \rho^{1-n} \int_{-\pi}^{\pi} e^{i(1-n)t} dt \quad \text{where } n \neq 1$$

$$\text{Hence, } = 0$$



We will not discuss the proof of CIT here. Cauchy proved this assuming  $f'(z)$  is cts (which is known to us which we know to be true now). This proof uses Green's Theorem & interested people can read it in Kreyszig. (Pg 654).

(Pg 659 Kreyszig) Consider  $\oint_C f(z) dz$  where  $C$  is unit circle oriented CCW.  
(Problem Set 14.2) In which of the cases is CIT applicable?

- (9)  $f(z) = e^{z^2}$ , (11)  $\frac{1}{4z-1}$ , (13)  $\frac{1}{z^4-1}$ , (14)  $\frac{1}{z}$

- (15)  $\operatorname{Re} z$ , (16)  $\frac{1}{11z-1}$ , (17)  $\frac{1}{|z|^2}$ , (18)  $\frac{1}{5z-1}$ , (12)  $(\bar{z})^3$ , (20)  $\log(1-z)$  with

$C$ : the boundary of the parallelogram with vertices  $\pm i, \pm(1+i)$ .

(9) ✓ 1) ✓  
\* (11) ✓ 2) No as  $z_0 = \frac{1}{4}$  is not a pt of analyticity of  $f(z)$  &  $z_0$  is in the interior of  $C$ .

(13) ✓ 3) ✓

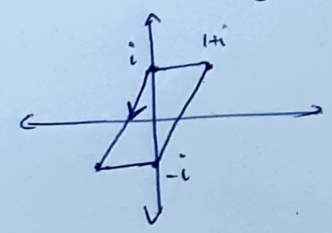
\* (14) ✓ 4) No, at  $z=0$ ,  $f(z)$  is not analytic

(15) ✓ 5) No (12) 9) No

16 6) No (20) 10)  $\log(1-z)$  is not analytic if  $1-z \leq 0$   
17 ✓ 7) No  $\Leftrightarrow z \geq 1$

18 8) No

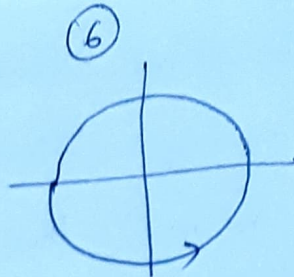
However,  $C$  is as given below



Since  $\log(1-z)$  is analytic on the interior of  $C$  & on  $C$ ,  $\oint_C \log(1-z) dz = 0$

$$I = \oint_C \frac{1}{4z-1} dz$$

$C$ : unit circle



$$z(t) = e^{it} \quad -\pi \leq t \leq \pi$$

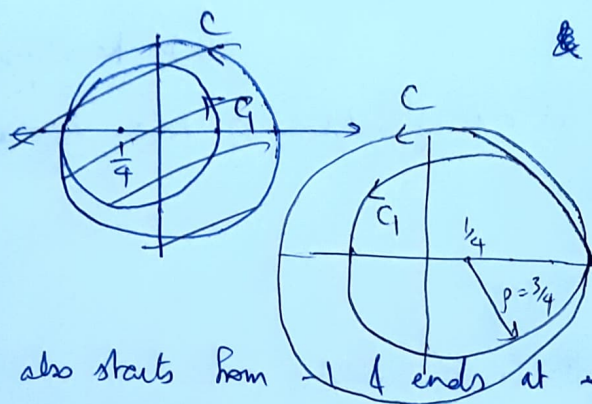
$$\int_{-\pi}^{\pi} \frac{1}{4e^{it}-1} i e^{it} dt = i \int_{-\pi}^{\pi} \frac{1}{4-e^{-it}} dt$$

$$= i \int_{-\pi}^{\pi} \left( \frac{1}{4-\cos t + i \sin t} \right) dt$$

One may multiply by conjugate & divide by conjugate, but it is lengthy.

Instead, write  $I = \oint_C \frac{1/4}{z-1/4} dz$

Now,



Consider  $C_1$

$$|z-1/4| = 3/4$$

$$z(t) = \frac{1}{4} + \frac{3}{4} e^{it} \quad -\pi \leq t \leq \pi$$

$C_1$  also starts from 1 & ends at 1.

$$\oint_{C_1} \frac{1}{z-1/4} dz = \oint_C \frac{1}{z-1/4} dz = \oint_C \frac{4}{4z-1} dz$$

$= 2\pi i$  from prev. problem by principle of deformation of path

Hence,  $I = \frac{1}{4} \times (2\pi i) = \frac{\pi i}{2}$ .

$$I = \oint_C \frac{1}{z} dz \quad ; \quad \text{Let } C: z(t) = e^{it}, \quad -\pi \leq t \leq \pi$$

$$= \int_{-\pi}^{\pi} e^{+it} i e^{it} dt = i \left. \frac{e^{2it}}{2i} \right|_{-\pi}^{\pi} = \frac{1}{2} \times [e^{2\pi i} - (e^{-2\pi i})] = 0$$



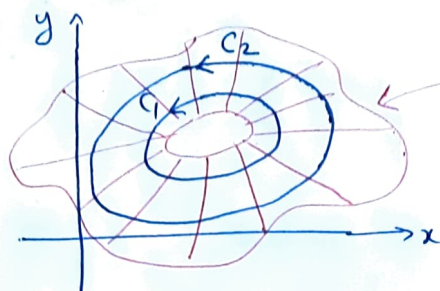
Let  $C_1$  &  $C_2$  denote positively oriented simple closed contours where

$C_1$  is interior to  $C_2$ . If a function  $f$  is analytic in the ~~closed~~ region on a doubly connected domain

D consisting of both these contours & all points between them, then

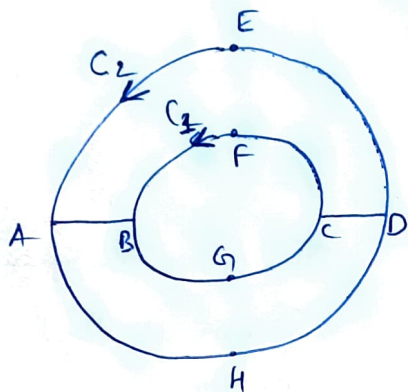
(Pg 159  
Churchill)  
8th edition

$$\int_{C_2} f(z) dz = \int_{C_1} f(z) dz$$



Note that red lines are dashed red lines meaning the boundary these lines are not part of the doubly connected domain.

Proof:-



Consider the simple closed contours

$$\Gamma_1 = \widehat{AB} + \widehat{BFC} + \widehat{CD} + \widehat{DEA}$$

&

$$\Gamma_2 = \widehat{AB} + \widehat{BGC} + \widehat{CD} + \widehat{DHA}$$

$$\int_{C_2} f(z) dz = \left( \int_{\widehat{DEA}} + \int_{\widehat{AHD}} \right) f(z) dz$$

$$\int_{C_1} f(z) dz = \left( \int_{\widehat{CFB}} + \int_{\widehat{BGC}} \right) f(z) dz$$

Note

$$\int_{\Gamma_1 + \Gamma_2} f(z) dz = \int_{\widehat{DEA}} + \int_{\widehat{AB}} + \int_{\widehat{BFC}} + \int_{\widehat{CD}} + \int_{\widehat{AB}} + \int_{\widehat{BGC}} + \int_{\widehat{CD}} + \int_{\widehat{DHA}}$$

$\int_{\Gamma_1} f(z) dz = \int_{\Gamma_2} f(z) dz$  as  $f$  is analytic on  $D$  &  $\Gamma_1$  &  $\Gamma_2$  are simple closed contours contained in  $D$ . And,

I can find a  $SCD$   $D_1 \subseteq D$  which contains  $\Gamma_1$  & another

$SCD$   $D_2 \subseteq D$  which contains  $\Gamma_2$ , such that  $f$  is analytic on  $D_1$  &  $D_2$



Thus,

$$\oint_{\Gamma_1} f(z) dz = \oint_{\Gamma_2} f(z) dz$$

~~Recall ~~Lemma~~,  $\int_{C_1} f(z) dz = \left( \int_{DEA} + \int_{AHD} \right) f(z) dz$  &  $\int_{C_1} f(z) dz =$~~

~~$0 = \oint_{\Gamma_1 + \Gamma_2} f(z) dz = 0 = \int_{AB} + \int_{BFC} + \int_{CD} + \int_{DEA} = \int_{AB} + \int_{BGC} + \int_{CD} + \int_{DHA}$~~

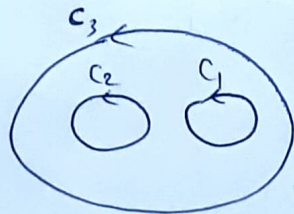
ie,  $\left( \int_{ADEA} - \int_{DHA} \right) = \left( \int_{BGC} - \int_{BFC} \right)$

$$\int_{\widehat{DEA} + \widehat{AHD}} (f(z) dz) = \int_{BGC + CFB}$$

$$\int_{C_2} f(z) dz = \int_{C_1} f(z) dz$$

This is also the Cauchy's Integral Theorem for doubly connected domain

Remark :- C17 for triply connected domain :-



$$\oint_{C_3} f(z) dz = \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz$$

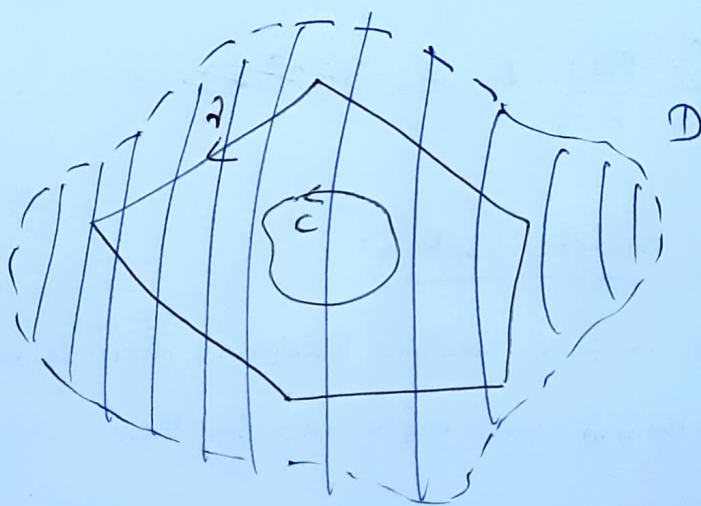
Conclusion :-

By principle of deformation of path, it follows that for  $m \neq -1$  &  $z \in D$

$$\oint_{\gamma} (z-z_0)^m dz = \begin{cases} 2\pi i & m = -1 \\ 0 & m \neq -1 \end{cases}$$

is true for any  $\gamma$  that is a simple closed contour enclosing  $z$ . We had seen the special case when  $\gamma$  is the circle  $|z-z_0| = \rho$ .

Take away :-  $\oint_C f(z) dz = \alpha$  for some  $C$  contained in a SCD  $D$  where  $f$  is analytic; As long as  $\gamma$  is a simple closed contour inside  $D$ , I can replace  $C$  by  $\gamma$  without any change in the integral.

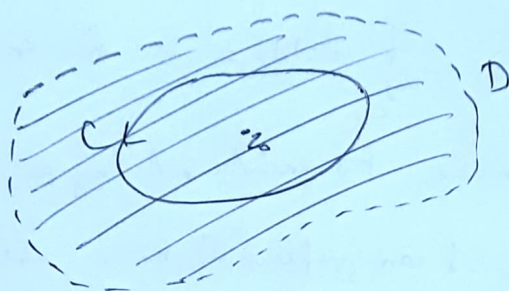
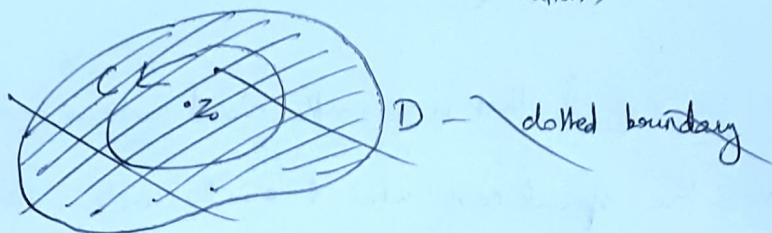


This ~~conclusion~~ is known as principle of deformation of paths since it tells us that if  $C_1$  is continuously deformed into  $C_2$ , always passing through points at which  $f$  is analytic, then the value of the integral never changes.



Theorem

Let  $f$  be analytic everywhere inside in a SCD  $D$ . Then for any point  $z_0 \in D$  & any simple closed contour  $C$  in  $D$  that encloses  $z_0$ ,  
(CCW orientation)



$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz = \cancel{\frac{1}{2\pi i} f(z_0)}$$

Derivatives of analytic functions

Recall that complex analytic functions have derivatives of all orders.

The following theorem gives expressions for them.

Theorem:

If  $f(z)$  is analytic in a domain  $D$ , then it has derivatives of all orders in  $D$ , which are also analytic functions in  $D$ . The values of these derivatives at a point  $z_0$  in  $D$  is given by

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \quad \text{More}$$

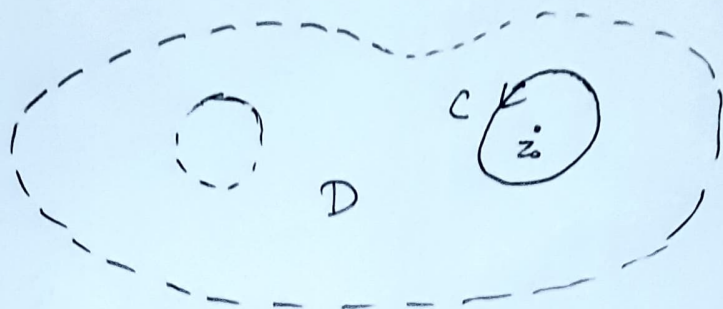
More generally,

$\forall n \geq 1$ ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

⑥ Here,  $C$  is any simple closed contour that encloses  $z_0$ , lies in  $D$  & whose full interior belongs to  $D$ , oriented CCW. (11)

No need for simply connected domain!



Eg. 1)  $\oint_C \frac{\cos z}{(z - \pi i)^2} dz$  where  $C$  is any simple closed contour enclosing  $\pi i$  (CCW)

$$= 2\pi i \times f'(\pi i) \quad \text{where } f(z) = \cos z$$

$$\text{Ans} = 2\pi i \times -\sin z \Big|_{z=\pi i} = 2\pi i \times -(\sin \pi i)$$

Eg. 2)  $\oint_C \frac{z^4 - 3z^2 + 6}{(z + i)^3} dz$  where  $C$  encloses  $-i$  (CCW)

$$f(z) = z^4 - 3z^2 + 6 \quad \& \quad f''(z) = \frac{d}{dz}(4z^3 - 6z) = 12z^2 - 6$$

$$f''(-i) = 12(-i)^2 - 6 = -18$$

$$\Rightarrow \frac{2!}{2\pi i} \times \oint_C \frac{f(z)}{[z - (-i)]^3} dz = f''(-i) = -18$$

$$\text{Ans} = \frac{-18 \times 2\pi i}{2} = -18\pi i$$

~~Cauchy's Inequality (S.D.T.)~~